MAJOR PROJECT · AI WEB APPLICATION

Smart Sugarcane AI

A website that helps a sugarcane farmer decide when to water the field, check plant disease from a photo, and pick the right variety and fertilizer.

smart-sugarcane-ai.vercel.app

7	12,000	13	0
features	rows of training data	pages	paid AI services used

Front end: React · Back end: Python (FastAPI) · Database: PostgreSQL

What the website does

Seven features. Each one asks a few simple questions, and gives an answer with the reason for it.

1 **Irrigation**

Tells you whether to water today, and how much

2 **Disease check**

Upload a leaf photo, get the likely disease

3 **Soil check**

Upload a soil photo, get the soil type

4 **Varieties**

Suggests which sugarcane variety suits your farm

5 **Fertilizer**

Tells you which nutrient the crop needs at this stage

6 **Assistant**

Ask a question in simple words, get an answer

7 **History**

Saves every result and shows if the crop is improving

! **Every answer says how it was made.**
Some features use a trained model, some use rules — the app always tells you which.

Front end — what the user sees

The part that runs inside the browser: the pages, the forms and the charts.

BUILT WITH

- **React** — builds the pages
- **TypeScript** — catches mistakes before they run
- **Tailwind CSS** — does the styling
- **Vite** — runs and builds the project

WHAT IS INSIDE

- 13 pages — one for each feature
- Login protection: you must sign in to use it
- Charts that show your past results
- Works on mobile and laptop, light and dark

HOW A USER MOVES THROUGH THE SITE

1. Open the website
2. Sign in to the account
3. Choose a feature
4. Type data or upload a photo
5. Get the answer with reasons

Back end — the brain on the server

The part the user does not see: it checks the input, does the calculation, saves it and sends the answer back.

BUILT WITH

- **Python** — the language everything is written in
- **FastAPI** — receives requests from the website
- **scikit-learn** — trains and runs the ML model
- **PostgreSQL** — stores users and all results

WHAT IT STORES

- Users — name, email, password (encrypted)
- Irrigation records — every watering decision
- Plant and soil results — every photo checked
- Chat messages — your questions and answers

WHAT HAPPENS WHEN YOU CLICK "GET RESULT"

1. The page sends your data to the server
2. The server checks every value is sensible
3. The model or the rules calculate the answer
4. The answer is saved and sent back to you

Which data we used

Three kinds of data go into this project — and one kind we could not get.

1 Irrigation data — for training

12,000 rows of example situations.
Each row has weather, soil and crop stage, and the correct water amount. The computer generated these rows using a standard water formula. This is what the ML model learns from.

2 Knowledge files — the book knowledge

4 files we wrote by hand, holding the farming knowledge:
12 sugarcane varieties, 7 diseases, 6 soil types and 5 growth stages. All the advice the app gives comes from here, so a farming expert can correct it without touching the code.

3 Photos — we could not get a good set

There is no reliable, correctly-labelled set of sugarcane disease photos available. So instead of faking a trained model, disease and soil use simple colour and texture rules, and every such result is clearly marked "DEMO" in the app.

4 Your own results — created as you use it

Every answer the app gives is saved to your account.
This becomes your own small dataset over time, and it is what the history chart and the assistant read from when they tell you whether the crop is improving.

Which method each feature uses

Only irrigation uses real machine learning. The rest use formulas and rules — and we say so.

#	Feature	Method	How it works
1	Irrigation	Machine Learning	Random Forest — it learned from the 12,000 examples how much water is needed
2	Water amount	Formula	Water needed = what the crop uses − the rain that fell. Standard agriculture method
3	Disease	Image rules (DEMO)	Measures how much of the leaf is brown, yellow or spotted, then matches a disease
4	Soil	Image rules (DEMO)	Matches the colour and roughness of the photo to the 5 common soil types
5	Varieties	Scoring	Gives each variety marks out of 100 — soil 30, region 24, water 20, climate 12, rest 14
6	Fertilizer & chat	Rules	Looks up the right advice for your growth stage from the knowledge files

✓ **Model result: 96 % accurate on the test data.** But we say this honestly — it is 96 % against the formula it learned from, not against a real field. That is why the DEMO features are labelled, not hidden.